Digestion in stomach

There are no enzymes present in the stomach to digest carbohydrates, despite the evidences citing the presence of gastric amylase, which is of lesser significance. Further, the low pH of stomach inactivates salivary amylase. The major digestive action of stomach on carbohydrates is that it helps in the cleavage of the glycosidic bonds present in sucrose. No further digestion of carbohydrates occurs in the stomach and the chyme passes into the duodenum.

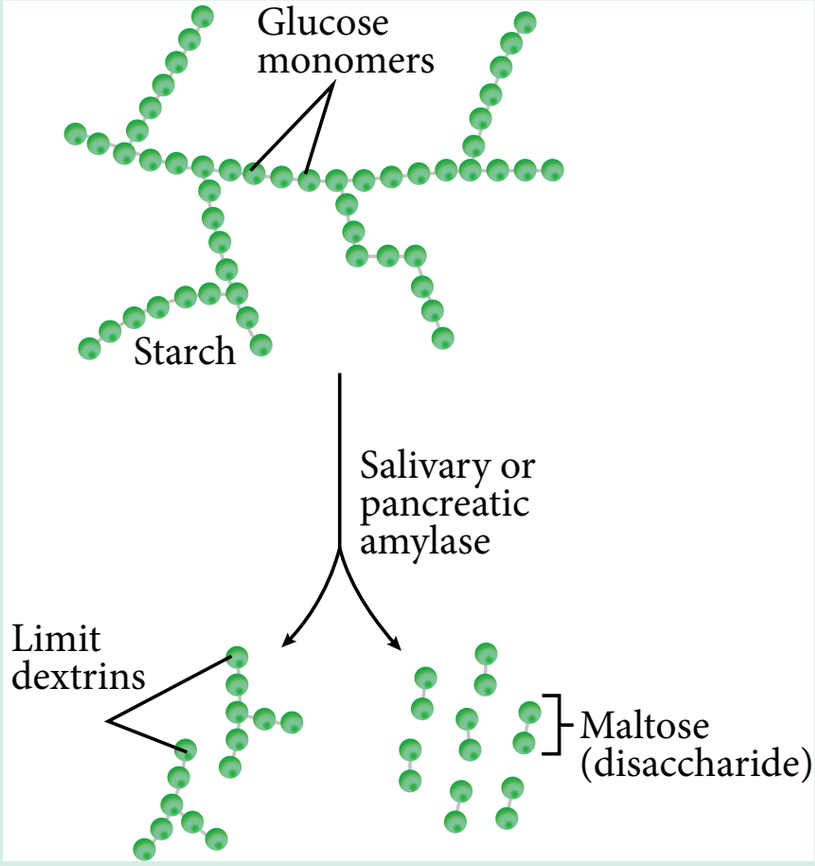


Figure 2.12 Action of amylases

Digestion in duodenum

The chyme on reaching the duodenum, gets mixed up with the bile juice and pancreatic juice that contains bicarbonate and gets neutralized. The pancreatic amylase, which is similar to the salivary amylase in enzymatic action, with optimum pH 6.9 to 7.1, acts upon the polysaccharides and breaks down the starch and glycogen to disaccharides like maltose and isomaltose (which contains α1→6 linkage)

Digestion in small intestine

Pancreatic amylase also requires chloride ion for its action and acts on the complex and partially digested carbohydrates. Pancreatic amylase completes the digestion of polysaccharides in the intestine because of

- its ability to break the interior linkages of complex carbohydrates
- the longer residual time of food in the duodenum and intestine

Moreover, pancreas secrete ten times more of pancreatic amylase into the intestinal lumen than that required to digest the carbohydrates in the diet.

Disaccharidases

There are four important disaccharidases in the human intestinal brush border membrane to hydrolyse the disaccharides (Table 2.1). They are:

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